

REPORT ON
BUSINESS PROCESS MAPPING
AT
**BPL MEDICAL
TECHNOLOGIES PVT. LTD.,
PALAKKAD**

Submitted to
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

In partial requirements for the award of the degree of
**Bachelor of Vocation in
BUSINESS PROCESS AND DATA ANALYTICS**

Submitted by
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Under the guidance of
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DDU KAUSHAL Kendra
COCHIN UNIVERSITY OF SCIENCE AND
TECHNOLOGY KOCHI-682022
JUNE 2025

CERTIFICATE FROM THE ORGANIZATION



May 27, 2025
BMTPL: HR: 03:05:25

CERTIFICATE

This is to certify that **Ms. Anjana Babu** (Reg No. 25123011), a student of the IV semester B. Voc at DDU Kaushal Kendra, CUSAT, Kochi, has successfully completed her 15-day project in our organization from **May 05, 2025, to May 23, 2025.**

We wish all the best in their future endeavors.

for BPL MEDICAL TECHNOLOGIES PRIVATE LIMITED

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This is to certify that this report “Business Process Mapping at **BPL MEDICAL TECHNOLOGIES PVT.LTD.**” is done as part of internship submitted to Cochin University of Science and Technology for the partial fulfilment of the award of the Degree of Bachelors of Vocation in Business Process and Data Analytics under the Faculty of Social Sciences, is the record of bonafide work done by Ms. Anjana Babu, Reg no: 25123011 under my supervision and guidance.

Dr.Vinod V Nair.

Assistant Professor, DDUKK

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Director, DDUKK

Kochi- 682022

Date:

DECLARATION

I, Anjana Babu, fourth semester B.Voc student of DDU KAUSHAL Kendra at Cochin University of Science and Technology do hereby declare that this REPORT ON BUSINESS PROCESS MAPPING AT BPL MEDICAL TECHNOLOGIES PVT.LTD., under the guidance of Dr. Vinod V Nair., Asst. Professor, DDU Kaushal Kendra in partial fulfilment of the requirement for the award of the degree of Bachelor of Vocation in Business Process and Data Analytics is a bonafide study I have done in the organization. I also declare that this report has not been previously formed the basis for the award of any Degree, Diploma, Associateship, Fellowship or another similar title of this or any other university or institution.

Place: Kochi

Date:

Anjana Babu

ACKNOWLEDGEMENT

I would like to thank God Almighty, for giving me the strength, patience and ability to complete the study and make this report on time.

I owe my thanks to Praveen P Nair, Asst. HR Manager of BPL MEDICAL TECHNOLOGIES PVT.LTD.,Palakkad for giving me permission and guiding me through the internship at BPL.

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ABSTRACT

The study is aimed at understanding the manufacturing processes of medical equipments. Conducted at BPL Medical Technologies Pvt. Ltd., the study involved interactions with team members to gather necessary data. Information was collected through interviews, observations, and document reviews. The gathered data from various sources were then classified, analyzed, and interpreted based on the critical processes identified. A comprehensive swim-lane diagram of these processes was created using Bizagi Modeler. This visual representation of the as-is process, allowing for clear identification and analysis. Additionally, a proposed 'to-be' process was mapped to suggest improvements. The study's systematic approach facilitated a thorough understanding of existing workflows and highlighted areas for enhancement, with the aim of optimizing efficiency and productivity in the manufacturing of BPL's medical products.

CHAPTER 1

INTRODUCTION

Business process mapping is a method used to visually represent the sequence of activities, tasks, and decisions that make up a specific business process. It provides a clear, structured view of how work flows through an organization, typically from the initial input to the final output.

By creating detailed diagrams, organizations can better understand the interactions between people, systems, and data at each step of the process. These maps help to clarify roles and responsibilities, uncover inefficiencies or bottlenecks, and identify areas for improvement or automation. Common types of business process maps include flowcharts, swimlane diagrams, and BPMN (Business Process Model and Notation), each offering varying levels of detail and structure.

Business process mapping not only supports process improvement initiatives but also promotes standardization, enhances communication among teams, and ensures that everyone involved has a shared understanding of how a process is supposed to function. In today's dynamic business environment, mapping processes is a key step toward achieving operational excellence and driving continuous improvement.

In the study conducted at BPL, the focus was on mapping the business process of manufacturing the Medical Technologies.

1.1 OBJECTIVES OF THE STUDY

- Identify and document key operational processes across departments to ensure a clear understanding of workflows.
- Highlight inefficiencies, redundancies, or delays in current processes that may impact product development, quality, or customer service.
- Standardize procedures to improve consistency, compliance, and quality assurance, especially in regulated environments like medical technology.
- Support process optimization and automation initiatives to increase efficiency and reduce manual workload.
- Enhance cross-functional communication by providing a visual representation of roles, responsibilities, and handoffs.
- Prepare for scalability by ensuring that processes can support growth, innovation, and global expansion.

1.2 SCOPE OF THE STUDY

The study is focused on key processes in the manufacturing of BPL products. The following product lines are covered:

- Nebulizer
- Patient Monitor
- Defibrillator

The study is done based on the details collected from the company via document review, personal and face to face interactions with the employees and managers and through observation of working and detailed study from the industry.

1.3 LIMITATIONS OF THE STUDY

- Due to confidentiality and proprietary restrictions, access to detailed internal documents, financial records, and sensitive operational data was limited.
- The process mapping is broad and may not cover in-depth workflows of every department within BPL Medical Technologies.
- Business processes in medical technology companies are subject to rapid changes due to regulatory updates and technological advancements, which may affect the long-term relevance of the findings.
- The duration of the study limited the extent of analysis and the possibility of performing detailed process simulations or ROI-based evaluations.

CHAPTER 2

INDUSTRY PROFILE

2.1 INDUSTRIAL OVERVIEW

MEDICAL TECHNOLOGIES INDUSTRY

The **medical technologies industry** encompasses a broad range of products and services designed to diagnose, monitor, and treat medical conditions. This includes devices such as imaging systems, diagnostic equipment, surgical instruments, patient monitoring tools, and life-support machines. The industry plays a critical role in improving patient outcomes, enhancing the efficiency of healthcare delivery, and supporting preventive and personalized medicine.

Driven by ongoing innovation, the medical technologies industry is marked by rapid advances in digital health, artificial intelligence, robotics, and minimally invasive procedures. Global demand continues to rise due to aging populations, the growing prevalence of chronic diseases, and increased healthcare access in emerging markets. At the same time, the industry must meet stringent regulatory standards, ensure product safety and effectiveness, and manage cost pressures from healthcare providers and governments.

Key trends include the integration of connected devices for remote patient monitoring, the adoption of data analytics for clinical decision support, and the push toward value-based healthcare. As competition intensifies and technology evolves, companies in the medical technologies space are focusing on R&D, strategic partnerships, and process optimization to stay ahead.

2.2 GLOBAL SCENARIO

The global medical technology (medtech) industry is a critical and rapidly expanding sector within healthcare, currently valued at around USD 570 billion and projected to surpass USD 800 billion by 2030. This growth is driven by several key factors, including an aging global population, a rising burden of chronic diseases like diabetes and cardiovascular conditions, and continuous technological advancements in diagnostics, treatment, and monitoring.

The industry covers a wide range of products and solutions, from traditional medical equipment such as imaging systems, surgical tools, and orthopedic implants, to advanced innovations like

AI-powered diagnostic devices, robotic surgery systems, and wearable health monitors. The increasing integration of digital technologies—including remote patient monitoring, telemedicine, and data-driven clinical decision-making—is reshaping how care is delivered globally.

In terms of regional dynamics, North America remains the largest market due to high healthcare expenditure and rapid adoption of new technologies. Europe follows with a strong focus on patient safety and compliance under new regulatory frameworks such as the EU MDR. Meanwhile, the Asia-Pacific region is emerging as the fastest-growing market, driven by expanding healthcare infrastructure, growing middle-class populations, and increased government spending, especially in countries like India and China.

Despite strong global momentum, the medtech industry faces several challenges. These include stringent regulatory approvals, cost pressures from healthcare providers and insurers, cybersecurity threats from connected devices, and supply chain vulnerabilities, which were highlighted during the COVID-19 pandemic.

Looking ahead, the future of the medtech industry is focused on digital health, personalized medicine, and value-based care. Companies are investing in smart, connected technologies that enable more precise, proactive, and patient-centered healthcare. As innovation continues and access to healthcare expands globally, the medical technology industry will remain a key enabler of improved health outcomes and more efficient healthcare systems.

Global Medical Technology Market Leaders:

1. Medtronic plc (Ireland/USA)
2. Johnson & Johnson MedTech (USA)
3. Siemens Healthineers (Germany)
4. GE HealthCare (USA)
5. Philips Healthcare (Netherlands)

2.3 INDIAN SCENARIO

The Indian medical technology (medtech) industry is emerging as a critical component of the country's healthcare ecosystem and is poised for significant growth in the coming years. Currently valued at around USD 11 billion, the industry is expected to reach USD 50 billion by 2030, driven by increased healthcare needs, growing disease burdens, rising incomes, and a stronger focus on healthcare accessibility and quality.

India's large and diverse population, with a rising incidence of non-communicable diseases like diabetes, heart disease, and cancer, is creating sustained demand for modern diagnostics and treatment devices. The expansion of hospitals, clinics, diagnostic labs, and telehealth platforms across both urban and rural areas is further propelling growth. The COVID-19 pandemic also acted as a catalyst for digital health adoption, accelerating the use of remote patient monitoring, wearable devices, and telemedicine platforms.

A key driver of the industry is the Indian government's push for self-reliance and local manufacturing under initiatives like Make in India, Ayushman Bharat (PM-JAY), and the Production-Linked Incentive (PLI) Scheme for Medical Devices. These policies aim to reduce dependency on imports—which currently account for over 75% of the market—and position India as a global hub for affordable and high-quality medical devices.

Despite its promising growth trajectory, the industry faces several challenges. The Indian medtech sector is still highly dependent on imports, especially for high-end imaging equipment, surgical tools, and implants. Additionally, the regulatory landscape is still evolving, with ongoing efforts to bring all classes of medical devices under a unified regulatory framework. Balancing innovation and affordability remains a challenge, particularly in making advanced technologies accessible to underserved rural populations.

The market is diverse and segmented into areas such as diagnostic imaging, in-vitro diagnostics, surgical instruments, patient monitoring, orthopedic devices, and disposables. Domestic players like BPL Medical Technologies, Trivitron Healthcare, Skanray Technologies, and Poly Medicure are increasingly gaining ground, while multinational companies like GE Healthcare, Philips, Siemens Healthineers, and Medtronic continue to have a strong presence through joint ventures and subsidiaries.

Looking ahead, the Indian medtech industry is expected to thrive on the back of increased investments, R&D, public-private partnerships, and a growing ecosystem of health-tech startups. As India improves its regulatory structure and manufacturing capabilities, it is well-positioned to serve not only its domestic population but also become a competitive exporter of medical devices to global markets.

Indian Medical Technology Market Leaders:

1. Healthium Medtech
2. Trivitron Healthcare
3. Meril Life Sciences
4. BPL Medical Technologies
5. Wipro GE Healthcare

CHAPTER 3

ORGANIZATIONAL PROFILE

3.1 BPL MEDICAL TECHNOLOGIES PVT. LTD.



Figure 3.1: BPL Palakkad building

BPL Medical Technologies, the Indian multinational, spearheads pioneering efforts in medical technology and innovation, keeping with legacy since 1967. Synonymous with trust, their products and systems are certified by national and international quality agencies. With their commitment to making in India, they help people experience great quality at unparalleled value. Their high standards of product performance have made them the no.1 choice of hospitals, clinics, and practising physicians across the country.

BPL Medical Technologies Private Limited is a trusted company that provides reliable medical equipment and dependable service to its customers. They have a large team of R&D that strides to design and develop medical devices as per the latest product features and offerings. They are also intensifying their efforts of incubating many new products. Their products manufactured in an ISO 13485: 2016 certified facilities located in Palakkad and Bangalore conform to global standards of quality assurance and best practices. BPL carries its legacy as the first Indian company to manufacture ECG devices in India.

Today, the company has evolved to be one of the largest Indian Medical Device companies, Their persistent efforts have been recognised when they were awarded with titles like ‘Most Leading Brand in Medical Equipment Company of the Year’ by Medgate Today in 2024, ‘Best

Healthcare Brand’ by Economic Times not once but twice, in the years 2023 and 2021. Furthermore, their testament to creating a workplace that inspires and nurtures talent at its core was solidified when they were conferred with the title of "Dream Company to Work in Medical Devices" at the Asia Pacific HRM Congress Awards 2021. So far in this journey, BPL MedTech has been recognized with the prestigious awards from Healthcare Excellence, Brand Trust

Report, Economic Times, Elets and countless more recognitions for their strength in customer centricity and medical technology innovation.

Their collaboration with the best brains across the world and their high-capacity manufacturing facilities ensure that their customers are equipped with the latest technologies at affordable prices. They will continue to introduce innovative products and expand their customer support network to enhance the productivity of healthcare.

With 14 offices across India, dedicated customer care team and network of over 500 professionals located in the farthest corners of the country, BPL is committed to provide their customers with service support - they are always within your reach.



Figure 3.2: Production plant

Vision

To be the preferred and trusted medical technology company, setting the highest standards on products and services, delivering affordable solutions to the last mile in the markets we serve, with commitment to customer centricity and ethical business practices.

Objectives:

- Design and developed, manufacturing solutions of quality, reliability and safety to meet or exceed requirements of our customers.
- Traded, co-branded, consumable products of quality, reliability and safety to meet or exceed the requirements of our customers.
- Manage risk during design, engineering and manufacture and through early warning, feedback and validation.
- Demonstrate standards of ethics and commitment to regulatory and safety standards.

- Continually improve the effectiveness and efficiency of products and processes across the organization.
- Enhance the competency and skills of our people through training and infrastructure.
- Co-operate with and assist our suppliers for continual improvement in products, services and practices.
- Implement system to track, analyze and handle customer complaints.
- Provide effective and responsive pre-sales, sales, post-sales and service operation.
- Drive customer satisfaction with all products and services and build long term loyalty by partnering with customers to help achieve their success.

Values

Our values form the basis of our culture and guide us in our everyday actions. They are the pillars on which our vision stands. Our purpose to ‘help save lives’ is based on the fundamental premise that ‘BMTPL ‘CARES’

CUSTOMER FOCUS

We believe in upholding our commitments, we strive for the best in quality in whatever we do, we drive on innovation, all of it, to create value for our customers - internal & external, in our pledge towards Customer Delight.

ACCOUNTABILITY

We are all Champions of our Company, driven by the same purpose to ‘Help Save Lives’, from taking ownership, to remaining committed from start to finish, in doing what is right, not just what is expected, and in going beyond, towards making a positive impact on everything we do.

RESPECT

We believe in listening, sharing, and being open to people, valuing their individual strengths and leveraging them. We believe in empowering our people with the right environment, efficient tools and opportune learning platforms towards nurturing a passionate team who in-turn respect the organization & its goals.

ETHICS

We uphold the ethics & values of BMTPL in every action and decision of ours. We pursue our goals with honour, integrity, trust, and fairness towards our employees, our customers, the business, and our Country.

SPEED

Challenging the status quo, racing against competition, by constantly striving for excellence, finding & discovering fast & smart solutions, towards continuous improvements and re-engineering self. We, therefore, are an organization driven by SPEED, of thought, of creation, of service, of delivering on our commitments & exceeding expectations.

Manufacturing Facilities

As a home-grown Indian brand, BMTPL is in alignment with the vision of our honourable Prime Minister's 'Atmanirbhar' Bharat (self-reliant India) in the medical device sector through the Make-in-India initiative. Our 13485 : 2016 and 9001 : 2015 certified state-of-the-art manufacturing facilities located in Bangalore and Palakkad adhere to the highest quality standards, ensuring that every product that bears the BPL Medical Technologies name is of the utmost quality and reliability. Synonymous with trust, our products are certified by national and international quality agencies, making them the preferred choice of hospitals, clinics and practicing physicians across the country. Their collaboration with the best brands across the world and their high-quality manufacturing facilities provide customers the latest technology at affordable prices.

Quality

They are focused on delivering products and services that consistently meet and exceed the customer's expectations in quality, reliability, safety and trust, by implementing and practicing an effective quality management system.

We are an ISO 13485:2016 Quality Management Systems certified Company. Most of their products are certified for European Safety Standard CE.

3.2 PRODUCT PROFILE

3.2.1 Defibrillators

Since launching its first defibrillator in 1975, BPL has been in the forefront in providing technological solutions for cardiac emergencies. It now offers a range of automatic as well as manual defibrillators that can help save lives.

DF 2509/R

With user-friendly features and simple use, DF-2509 monophasic defibrillator comes with a 5.7 inch LCD screen and has synchronized cardioversion, remote charging through paddles as well as event storage facility.

Features

- Monophasic defibrillator
- 5.7-inch LCD screen
- User-selectable energy levels from 2J to 360J
- Synchronized cardioversion
- Less than 5 seconds to charge to 360J
- Remote charging from paddles
- 24 events memory
- Minimum 100 discharge of 360J with fully charged new battery



Figure 3.3: DF 2509/R

3.2.2 Patient Monitor

Reliable patient monitoring is very important during critical care. BPL Medical offers an array of high-quality and affordable patient monitors for your practice, developed to get you the correct data you require easily, securely and quickly. Backed by our strong network of professionals, we are committed to providing you with uninterrupted & seamless support at all times.

Vivid Vue 12

Reliable patient monitor for OT and critical care environment, Vivid Vue 12 comes with a 12.1" high resolution display with an optional touchscreen feature and allows monitoring advanced parameters such as IBP, CO2, Gas monitoring. It has integrated HL7 communication protocol with HIS connectivity.

Features

- 12.1" high-resolution display (optional touchscreen)
- Standard configuration - ECG, RESP, SpO2 (MASIMO / Nellcor / BPL OxySat+), NiBP (SunTech), TEMP (optional 12-lead ECG)
- Optional parameters - Dual IBP, ETCO2 (Masimo Sensors), Anesthesia Gas Monitoring (Masimo Sensors)
- Enhanced connectivity - HL7 with HIS connectivity, USB for patient data output, DVI for external (Slave) display, Nurse call function. Optional: Medius+ central nursing station with remote monitoring feature
- Convenient feature of standby, night mode & snapshot (screenshot)
- Comprehensive data storage available
- High performance Li-ion battery, providing upto 3.5 hrs backup time in low power consumption



Figure 3.4: Vivid vue 12

3.2.3 Nebulizer

According to the Global Asthma Report, approximately 35 million people in India suffer from asthma. With the levels of indoor allergens constantly increasing, the prevalence of asthma is constantly on the rise. While there is no known cure, it definitely can be controlled by breathing therapy and medications. In order to help children inhale, nebulizers turn the liquid medicine into a mist that is breathable.

BPL Breathe Ezee N8

Breathe Ezee N8 compressor nebulizer is equipped with high-quality compressor for enhanced performance and high product life. Its inbuilt fuse ensures safe operation in electricity fluctuations. The device comes with an adult and paediatric mask, making it suitable for all age groups.

Features

- **Efficient nebulization:** It combines a short inhalation time with efficient medication delivery.
- **Superfine particles:** Breaks down the medication to super fine particles, ideal for reaching the lower airways assuring effective treatment
- **Low noise operation:** An efficient compressor ensures minimal noise while operation, producing noise of less than 60dB
- **One switch operation:** To start and end therapy with a click of a button makes it easy to operate
- **Ergonomic design:** Light-weight compact design with a handle that facilitates easy portability



Figure 3.5: BPL Breathe Ezee N8

3.3 ORGANIZATIONAL STRUCTURE AND DEPARTMENT ANALYSIS

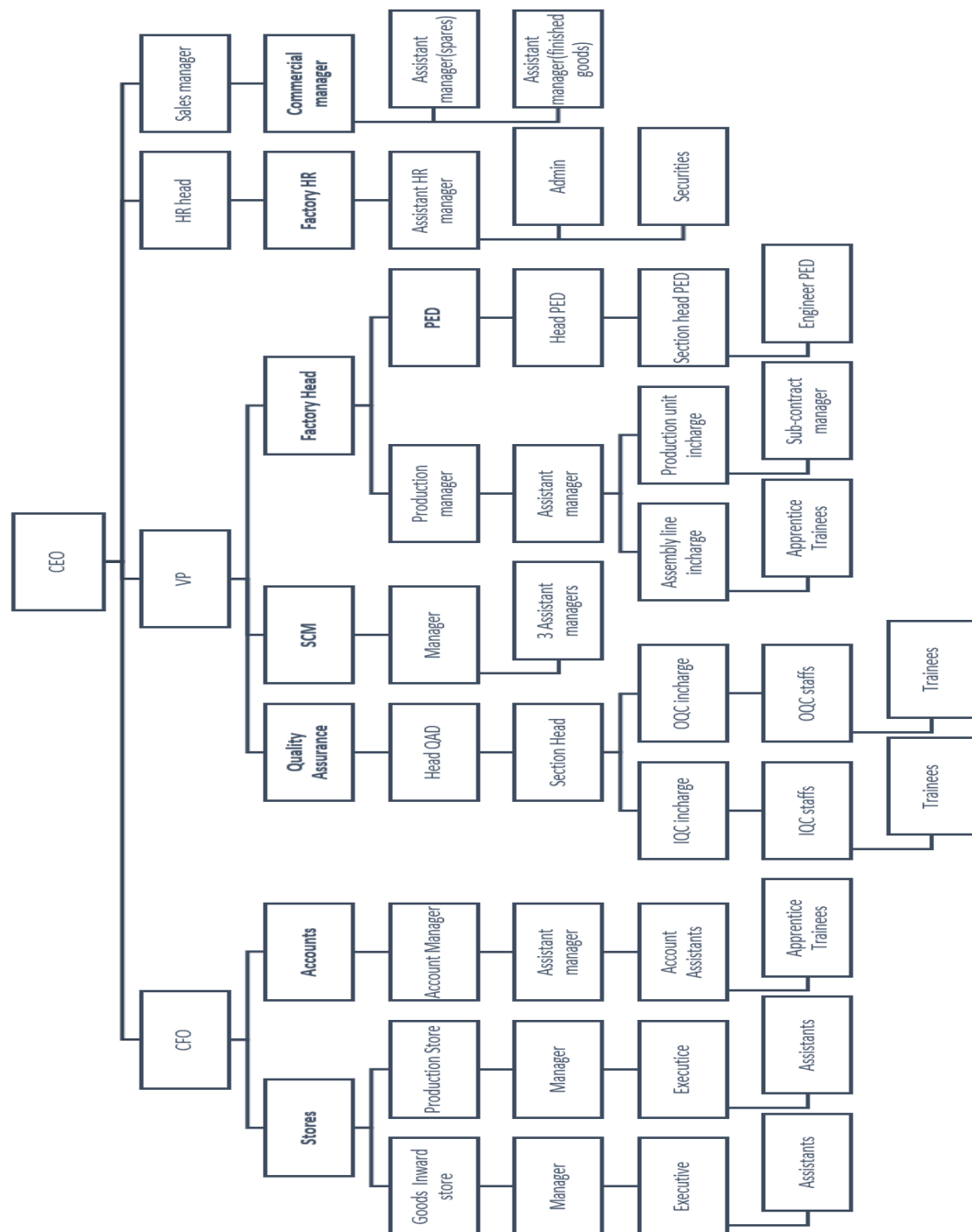


Figure 3.6: Organizational structure of BPL

DEPARTMENTS

1. Production Engineering Department
2. Human Resources (HR)
3. Production
4. Supply Chain Management (SCM)
5. Accounting and Finance
6. Commercial
7. Stores (Inventory Management)
8. Quality Assurance (QA)

Here's a breakdown of the functions of each department in a Medical Technologies Company, assuming it manufactures or distributes medical devices, diagnostics, or similar products:

1. Production Engineering Department

The Production and Engineering Department is responsible for the assembly, testing, and continuous improvement of medical devices. It ensures that all products are built to high-quality standards, using precise engineering processes, optimized workflows, and compliant equipment. The team also drives process innovation, supports technical troubleshooting, and collaborates with quality and regulatory teams to maintain safety and compliance.

Functions:

- Design and optimize manufacturing processes for medical devices.
- Ensure compliance with regulatory standards (e.g., FDA, ISO 13485).
- Implement automation and quality improvements.
- Collaborate with R&D and QA to scale up new products.
- Perform root cause analysis and corrective actions for production issues.

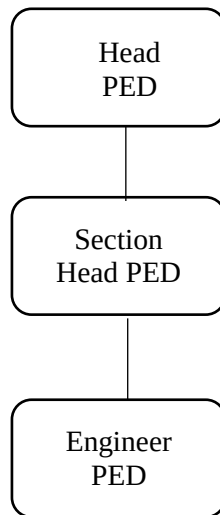


Figure 3.7: Structure of Production & Engineering dept.

2. Human Resources (HR) department

The HR Department manages talent acquisition, employee development, and organizational culture to support the company's mission in delivering quality medical technologies. It ensures compliance with labor laws, drives training and performance management, and fosters a safe, inclusive, and productive workplace aligned with industry standards.

Function:

- Talent acquisition and recruitment of skilled staff (engineers, technicians, regulatory experts).
- Training and development, especially on safety and compliance.
- Payroll, employee relations, and benefits administration.
- Ensuring workplace safety and regulatory compliance.
- Performance management and succession planning.

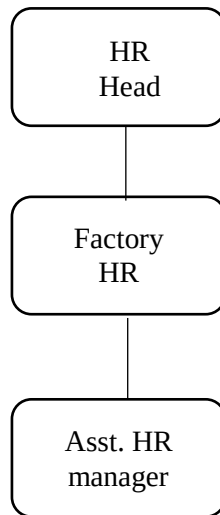


Figure 3.8: Structure of HR dept.

3. Production department

The Production Department is responsible for the assembly and output of medical devices, ensuring products meet quality, safety, and regulatory standards. It focuses on efficient workflows, strict adherence to SOPs, and on-time delivery, supporting the company's commitment to reliable and compliant healthcare solutions.

Function:

- Execute daily manufacturing of medical devices or components.
- Monitor production lines and ensure efficiency.
- Maintain production schedules and meet output targets.
- Follow Good Manufacturing Practices (GMP).
- Coordinate with QA and engineering for quality checks and improvements.

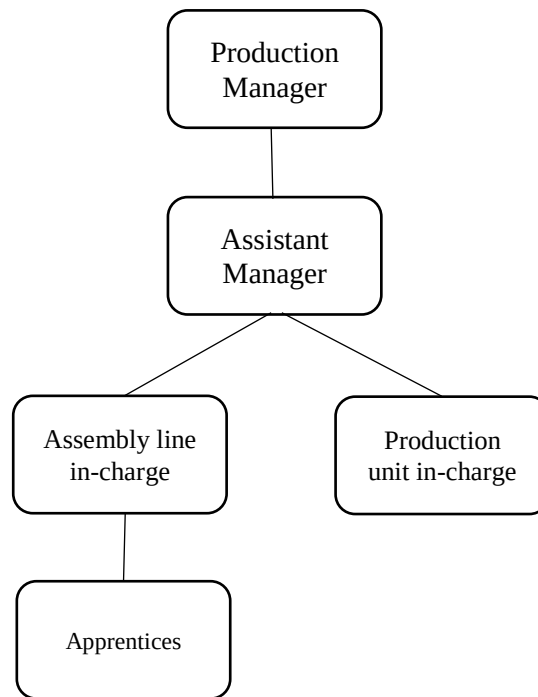


Figure 3.9: Structure of Production dept.

4. Supply Chain Management (SCM) department

The SCM Department oversees the procurement, logistics, and inventory management of components required for medical device assembly. It ensures timely availability of quality materials, optimizes vendor relationships, and maintains cost-effective, compliant, and traceable supply chain operations to support uninterrupted production.

Function:

- Manage procurement of raw materials and components.
- Vendor management and strategic sourcing.
- Oversee logistics, warehousing, and inventory levels.
- Forecast demand and plan production material requirements.
- Ensure timely delivery while minimizing costs and delays.

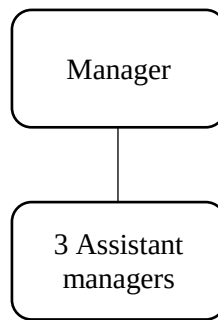


Figure 3.10: Structure of SCM dept.

5. Accounting and Finance department

The Accounting and Finance Department manages the company's financial planning, budgeting, accounting, and reporting. It ensures regulatory compliance, oversees cost control, and provides financial insights to support strategic decisions, enabling sustainable growth and operational efficiency in the medical technology sector.

Function:

- Budgeting, forecasting, and financial reporting.
- Cost control and pricing strategy support.
- Managing accounts payable/receivable.
- Support audits (internal and external).
- Evaluate financial risks and investment opportunities.

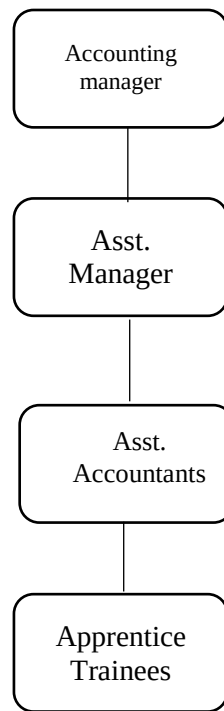


Figure 3.11: Structure of accounting & finance dept.

6. Commercial department

The Commercial Department drives the company's sales, marketing, and customer relationship strategies. It focuses on market growth, revenue generation, and ensuring that products meet customer needs and regulatory expectations, helping the company expand its presence in the medical technology industry.

Function:

- Market analysis and product positioning.
- Promote medical devices to hospitals, clinics, and distributors.
- Manage customer relationships and key accounts.
- Ensure compliance with advertising and promotional regulations.
- Drive revenue growth through campaigns and partnerships.

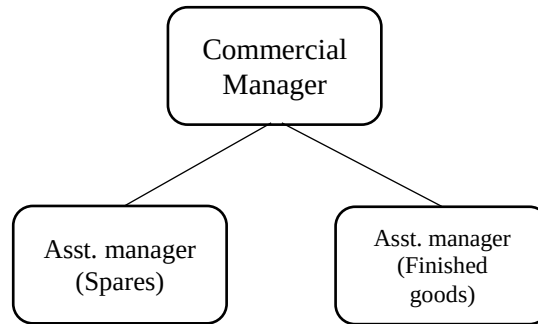


Figure 3.12: Structure of Commercial dept.

7. Stores (Inventory Management)

The Stores Department is responsible for the receipt, storage, and issuance of components and raw materials. It ensures accurate inventory management, maintains proper handling and traceability of medical-grade materials, and supports smooth and compliant assembly operations.

Function:

- Maintain and organize inventory of raw materials, components, and finished goods.
- Track stock levels and issue materials to production.
- Ensure proper storage conditions (especially for temperature-sensitive items).
- Implement FIFO/LIFO and inventory control methods.
- Perform regular stock audits and reconciliations.

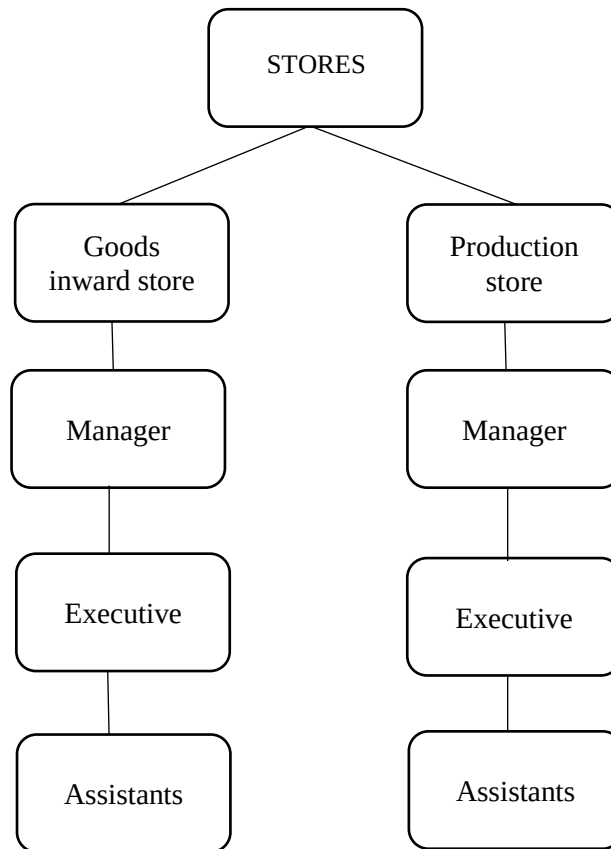


Figure 3.13: Structure of stores

8. Quality Assurance (QA) department

The QA Department ensures that all medical devices meet regulatory, safety, and quality standards. It oversees process validation, documentation, inspections, and audits, ensuring every product delivered is safe, effective, and compliant with industry regulations.

Function:

- Establish and enforce quality standards and SOPs.
- Conduct internal audits and manage external regulatory inspections (FDA, CE, etc.).
- Ensure compliance with ISO 13485 and other applicable standards.
- Investigate and resolve non-conformances and customer complaints.
- Validate equipment, processes, and final products before release.

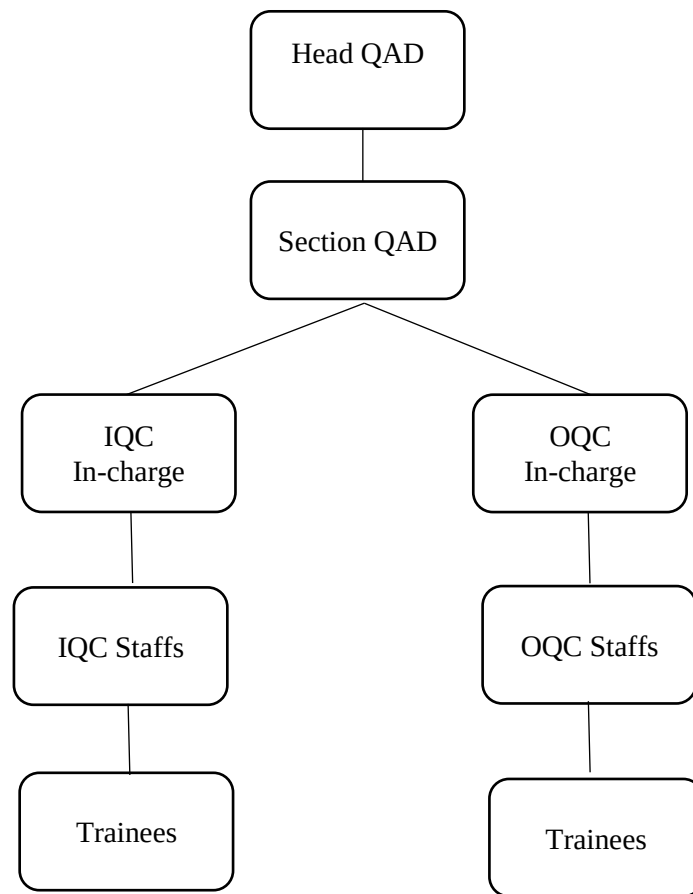


Figure 3.14: Structure of QA dept.

CHAPTER 4

PROCESS MAPPING

4.1 PROCESS IDENTIFICATION

Process identification systematically organizes an identified process within an organization. It aids in understanding issues within the process and adopting improvement techniques.

The diagram below illustrates the process in the overall organization:

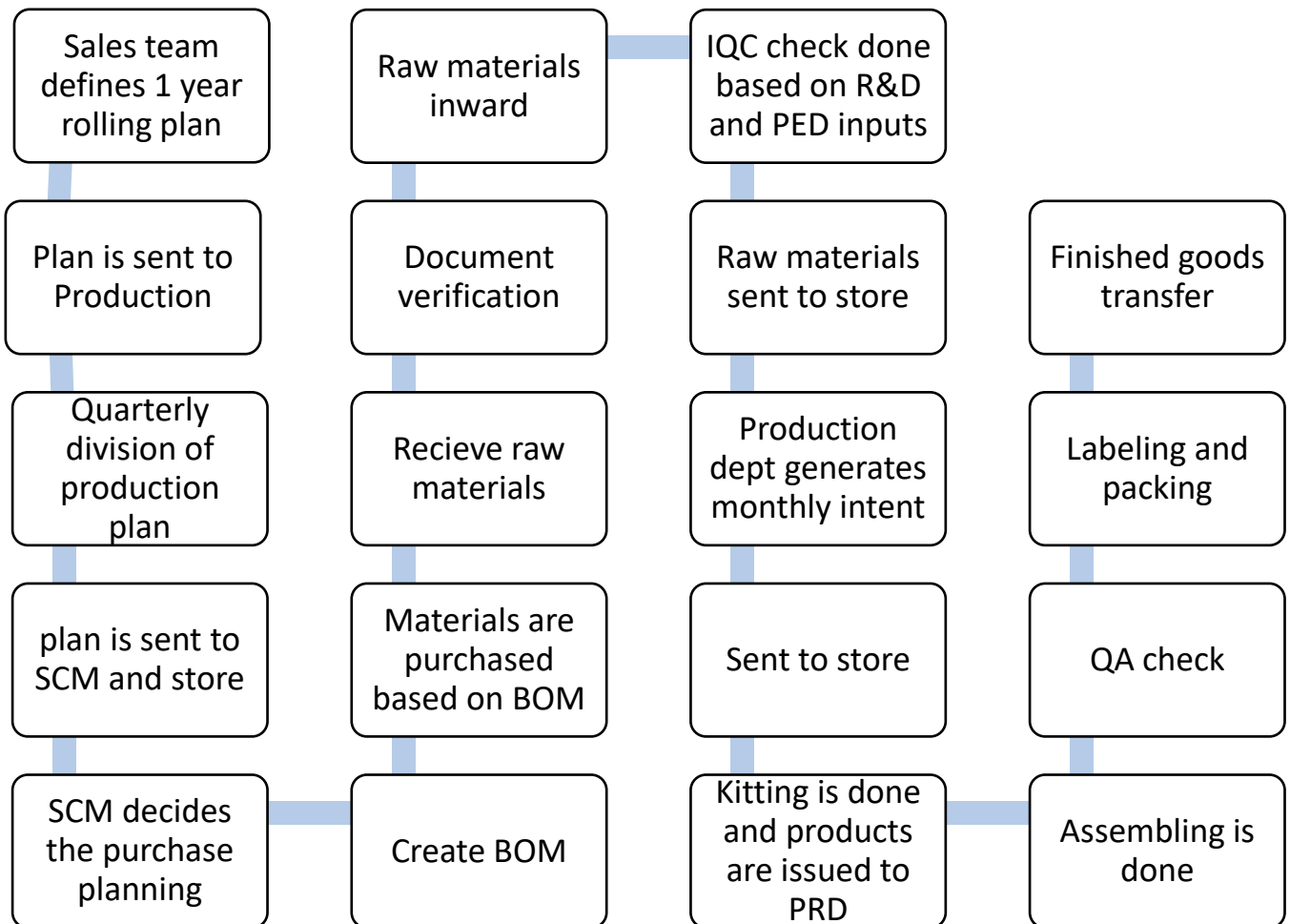


Figure 4.1: Overall process in the organization

The diagram below illustrates the process in the production department:

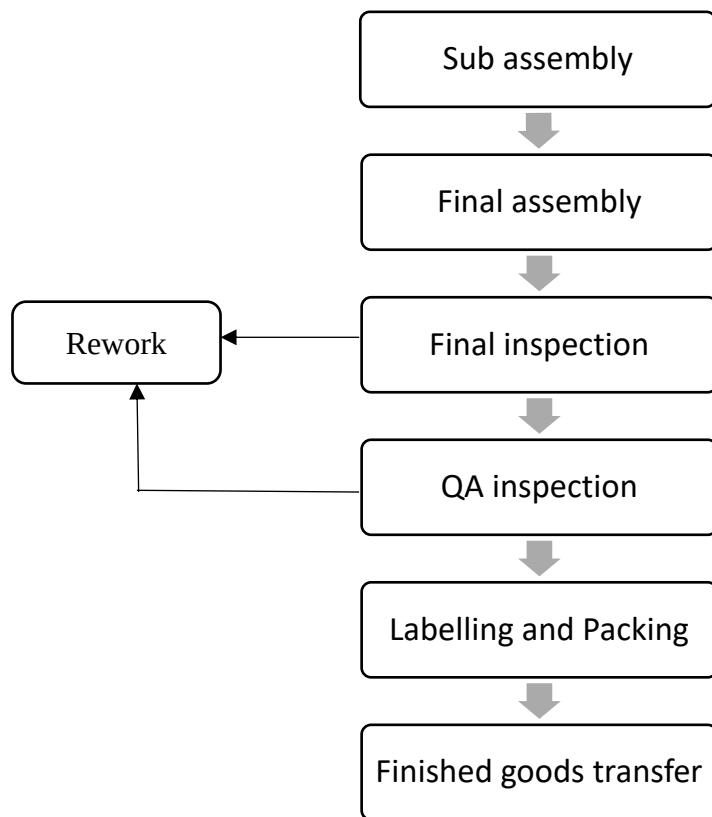


Figure 4.2: Process of production department

1. Sub-Assembly

Explanation: Involves assembling smaller components or modules that will be part of the final product.

Example: For a patient monitor, this might include assembling the circuit boards, display screens, and sensor units separately.

Purpose: Ensures each module is functional before full integration, allowing for focused testing and efficiency.

2. Final Assembly

Explanation: All sub-assembled modules are integrated into the final medical device.

Example: Combining the display unit, internal electronics, outer casing, and user interface of an infusion pump.

Purpose: To complete the full product according to specifications, preparing it for testing and inspection.

3. Final Inspection

Explanation: A comprehensive visual and functional check of the fully assembled medical equipment.

Example: Inspecting a nebulizer for missing parts, correct labeling, cable connections, and physical damage.

Purpose: Ensures the product is built correctly and meets design and safety standards before advanced quality checks.

4. Quality Assurance (QA) Inspection

Explanation: A detailed inspection by QA personnel to verify compliance with regulatory standards (e.g., FDA, ISO 13485).

Example: Testing a defibrillator's response time, energy delivery, software performance, and calibration.

Purpose: Ensures the device is safe, effective, and reliable for clinical use.

5. Labelling and Packing

Explanation: The device is labeled with serial numbers, safety warnings, user instructions, and regulatory compliance marks. Then it's securely packaged.

Example: A diagnostic device might be packed with user manuals, power cables, probes, and CE marking labels.

Purpose: To provide all required information to users and to ensure the product is protected during transportation.

6. Finished Goods Transfer

Explanation: The final, packaged products are moved to the finished goods inventory or shipping department.

Example: Transferring boxed defibrillators to the distribution warehouse.

Purpose: Marks the product as ready for delivery to hospitals, clinics, or distributors.

4.1.2 PROCESS IDENTIFICATION OF PRODUCT LINES:

1. NEBULIZER

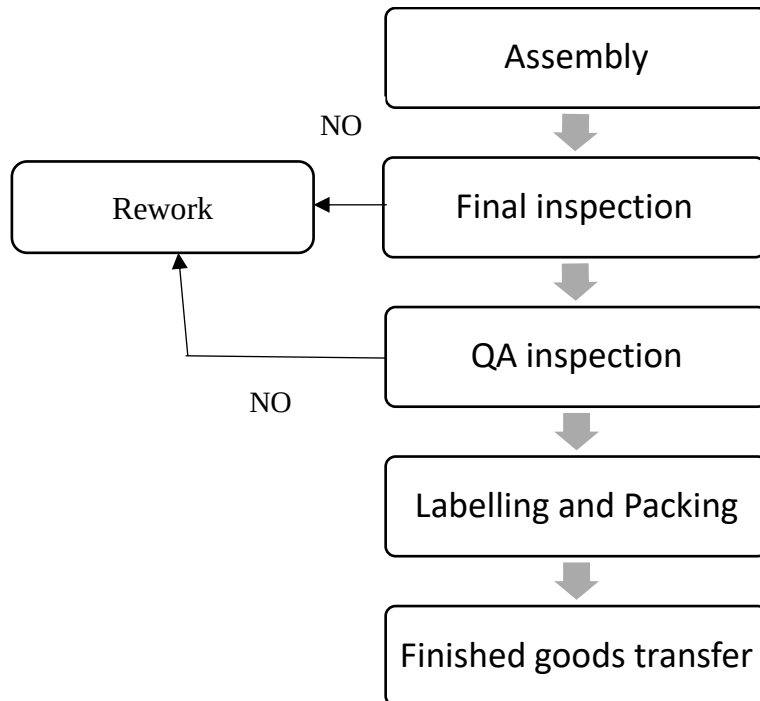


Figure 4.3: Process of Nebulizer

2. PATIENT MONITOR

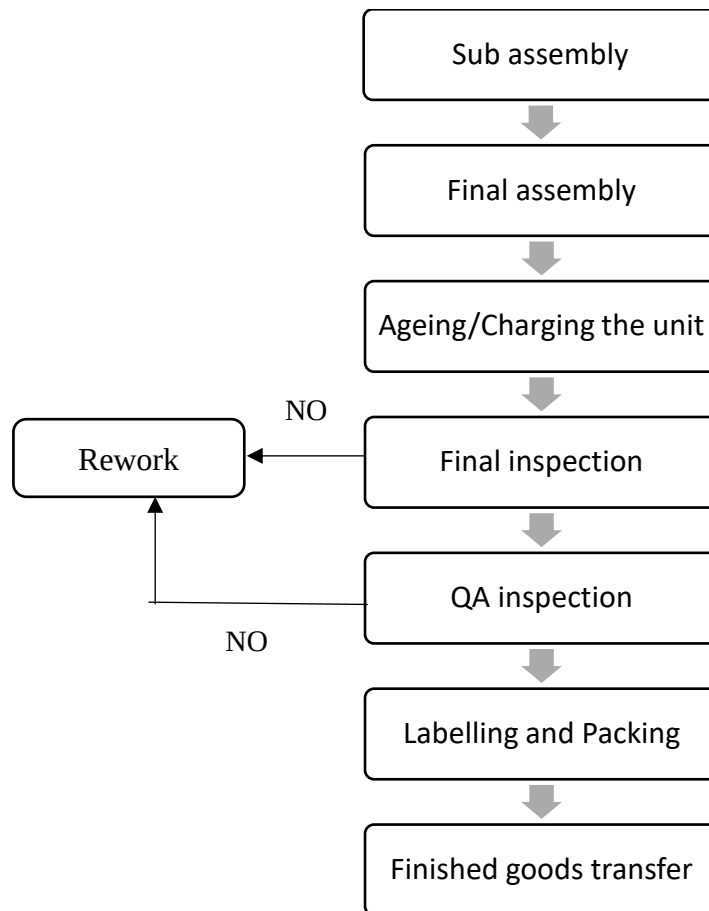


Figure 4.4: Process of Patient monitor

3. DEFIBRILLATOR

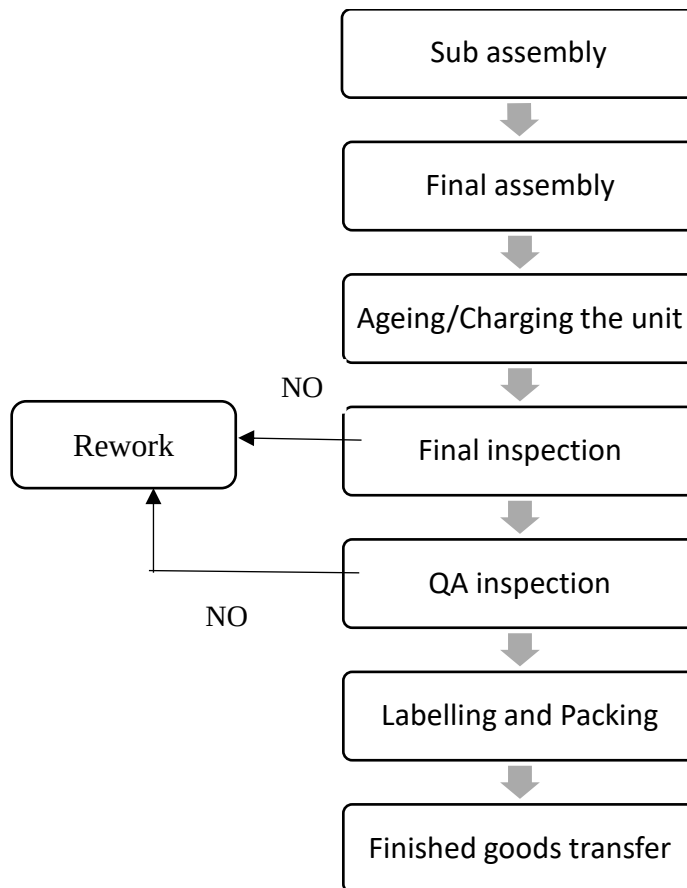


Figure 4.5: Process of Defibrillator

4.2 PROCESS DISCOVERY

Process discovery is a technique used to automatically identify, visualize, and understand real-world workflows within an organization by analyzing data from system logs or user activities. It helps uncover deviations, bottlenecks, and inefficiencies compared to documented processes. This insight enables businesses to optimize operations, ensure compliance, and make informed decisions about process improvements. Process discovery is a key component of process mining and is widely used in business process management to bridge the gap between theoretical processes and actual execution.

4.2.1 Primary Methods:

The primary data collection method is the collection of data directly from its sources. Like:

- Questionnaire
- Face to face interaction
- Personal interviews
- Observation

4.2.2 Secondary Method:

The Secondary research is a systematic investigation in which the researcher depends on existing data in the course of the research process. Like:

- Website of the organization
- Yearly plan of Operation book
- Previous reports of the organization

4.3 **AS – IS PROCESS**

An **As-Is process** represents the current state of a business process within an organization. It involves a detailed analysis and documentation of how a process is presently being carried out — including the sequence of activities, the roles and responsibilities involved, the tools or systems used, and the interactions between different departments or stakeholders.

The main goal of mapping the As-Is process is to gain a clear understanding of existing workflows, identify inefficiencies, bottlenecks, redundancies, or areas lacking standardization. This analysis serves as the foundation for proposing improvements or designing a more efficient To-Be process.

In the context of a company like BPL Medical Technologies, studying the As-Is process helps in uncovering operational challenges, compliance gaps, or workflow delays that may affect product development, regulatory approvals, or customer satisfaction.

Figure 4.6: as-is process

CHAPTER 5

PROCESS ANALYSIS

5.1 BUSINESS PROCESS ANALYSIS:

Business Process Analysis (BPA) is a methodical approach to examining, understanding, and evaluating the various processes within an organization. It involves breaking down each step of a process to identify how work is performed, who performs it, what tools or systems are used, and what inputs and outputs are involved.

The primary objective of BPA is to identify inefficiencies, redundancies, or bottlenecks and uncover opportunities for improvement, automation, or cost reduction. By analyzing these processes, organizations can enhance productivity, improve quality, ensure compliance, and better align operations with strategic goals.

In a company like BPL Medical Technologies, business process analysis can be crucial in optimizing workflows related to manufacturing, quality control, regulatory compliance, and customer service — ensuring efficient delivery of high-quality medical devices to the market.

5.1.1 Quantitative Process Analysis:

Quantitative Process Analysis is the evaluation of business processes using measurable data and numerical metrics. It focuses on analysing aspects such as process time, cost, error rates, throughput, and resource utilization to assess performance and identify areas for improvement.

This method relies on hard data rather than subjective observations, enabling organizations to make data-driven decisions and track the impact of process changes over time.

The Different types of Quantitative Process Analysis include:

- Lean management theory
- Time analysis
- Six sigma
- Error/Defect rate analysis

5.1.2 Qualitative Process Analysis:

Qualitative Process Analysis is the study of business processes based on non-numerical information such as observations, interviews, feedback, and documentation. It focuses on

understanding the why and how behind process behaviours, user experiences, communication patterns, and organizational culture.

Rather than relying on data and statistics, this approach emphasizes contextual insights, identifying root causes of issues, and exploring improvement opportunities that may not be visible through numbers alone.

The different types of Qualitative process analysis includes:

- Organizational behavior theory
- Root-cause analysis
- Interview based analysis
- Issue register

5.2 FLOW ANALYSIS

Flow Analysis is the examination of how work, information, or materials move through a process or system. It helps identify the sequence of steps, decision points, delays, and handoffs, aiming to improve efficiency, reduce waste, and streamline operations.

In business, flow analysis is commonly used to visualize and optimize workflows, often through tools like flowcharts, process maps, or value stream maps.

5.2.1 Cycle Time

Cycle Time refers to the total amount of time required to complete one full cycle of a specific process, starting from the initiation of the task to its final completion. It includes all the time spent actively working on the task, as well as any delays, waiting periods, inspections, or handoffs that occur during the process.

Cycle time is a key performance indicator (KPI) used in manufacturing, service delivery, and business operations to assess process efficiency, identify bottlenecks, and support continuous improvement efforts.

Cycle time includes both value-added Times as well as non-value-added Times.

$$\text{Cycle Time} = \text{Finish Time} - \text{Start Time}$$

$$\text{Cycle Time Ratio} = (\text{Finish Time} - \text{Start Time}) / \text{Units Produced}$$

5.2.2 Cycle Time Efficiency

Cycle Time Efficiency (CTE) is a performance metric that measures the proportion of time spent on value-added activities during a process compared to the total cycle time. It indicates how efficiently a process converts input into output without unnecessary delays or waste.

It is calculated using the formula:

$$\text{Cycle Time Efficiency} = (\text{Value-Added Time} / \text{Cycle Time}) * 100$$

- Value-Added Time is the time spent directly contributing to the final product or service.
- Total Cycle Time includes both value-added and non-value-added activities (like waiting, rework, or idle time).

5.2.3 Lead Time

Lead Time refers to the total duration between the initiation of a process and its final completion or delivery. It encompasses every step involved — including order placement, processing, manufacturing, inspections, packaging, and shipping. Lead time includes both value-added and non-value-added activities such as delays, waiting periods, or approvals.

Lead time is a critical metric in supply chain management, production, and service delivery, as it directly impacts customer satisfaction, inventory management, and overall operational efficiency.

$$\text{Lead Time} = \text{Value-Add Time} + \text{Non-Value-Add Time}$$

5.3 CYCLE TIME OF PRODUCTS:

- Lead time for **Nebulizer** = 3 minutes(Component collection and inspection)+ 5 minutes(base unit set-up)+ 6 minutes(wiring & PCB)+ 4 minutes(Enclosure assembly)+ 3 minutes(External fittings)+ 6 minutes(final QA inspection)+ 2 minutes(labeling)+ 4 minutes(packing)+ 5 minutes(delay in between the processes)
= **38 minutes**
- Value-added time = 24 minutes
Cycle time efficiency = $(24/38)*100 = 63\%$

5.4 ISSUE REGISTER

I am utilizing Issue Register Analysis, a qualitative process analysis technique.

An Issue Register is a formal document or tool used to record, track, and manage issues that arise during a business process or project lifecycle. It captures key information about each issue, such as its description, impact, status, responsible person, and corrective actions.

It serves as a centralized log that ensures all issues are acknowledged, assigned, monitored, and resolved in a systematic and transparent manner.

Issue Name	Explanation	Assumption	Qualitative Impact	Quantitative Impact	Improvement Actions
Equipment Unavailability	Manual tools shared between stations or not working	No tool maintenance schedule	Queue at stations, slow progress	15–20 mins loss per batch	Allocate tools per station, introduce preventive tool maintenance
Lack of Real-Time Monitoring	No visibility into live status of WIP and delays	No digital tracking system	No proactive intervention; issues escalate	Production loss: 10 units/week	Use ERP or MES dashboards with alerts for bottlenecks
Rework Loops	Frequent errors leading to repeated correction cycles	Poor training or tool slippage	Wasted time, fatigue, more delays	5–7% increase in overall lead time	Provide refresher training, use jigs/fixtures to reduce error
Long QC Approval Cycles	Delay in quality inspection between steps or at end	Limited QC staff	Blocking flow of work, waiting time	Avg. delay: 10 - 15 mins/unit	Use tablet-based digital checklists or camera-based auto-inspection

Table 5.1: Issue Register

CHAPTER 6

PROCESS IMPROVEMENT

6.1 QUALITY IMPROVEMENT TECHNIQUES:

Quality Improvement Techniques are structured methods and strategies used to enhance the performance, reliability, and effectiveness of processes, products, or services within an organization. These techniques aim to reduce defects, minimize waste, increase efficiency, and ensure that outputs consistently meet customer expectations and regulatory standards.

They are widely used across industries, including healthcare and medical technology, to drive continuous improvement and operational excellence. By identifying root causes of problems, streamlining workflows, and applying data-driven decision-making, quality improvement techniques help organizations like BPL Medical Technologies maintain high standards of product safety, compliance, and customer satisfaction.

Common techniques include tools like the PDCA Cycle, Six Sigma, Lean Principles, Fishbone Diagrams, and Control Charts, all of which support a culture of continuous improvement and innovation.

The Quality improvement technique adopted in the process analysis is the ‘FISHBONE DIAGRAM’.

6.1.1 FISHBONE DIAGRAM:

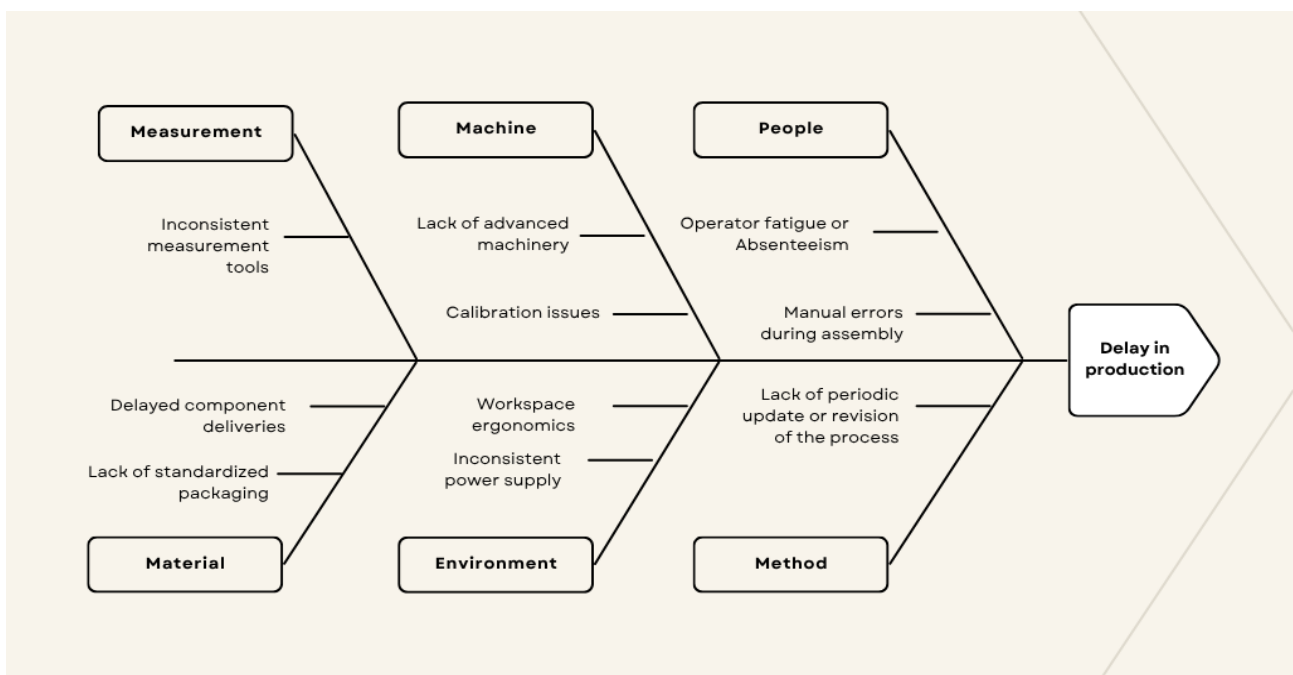


Figure 6.1: Fishbone diagram

6.2 IMPROVEMENTS:

- Establish a centralized ERP system for real-time monitoring and updation across all departments.
- Standardize packaging materials to avoid inconsistencies.
- Improve workspace ergonomics to reduce operator fatigue and error.
- Introduce automation in production
Eg: Automated pick and place tools for assembly and automated labelling and packing
- Digitalize SOPs and protocols

6.3 TO-BE PROCESS:

The To-Be Process represents the ideal or improved future state of a business process, redesigned to be more efficient, cost-effective, and aligned with organizational goals. It is developed after analyzing the current (As-Is) process and identifying gaps, bottlenecks, or inefficiencies. The To-Be process incorporates best practices, technology enhancements, and quality improvements to deliver better performance and compliance.

Figure 6.2: to-be process

CHAPTER 7

CONCLUSION

7.1 FINDINGS

1. Strong Commitment to Quality
2. The organization places a high priority on product quality and compliance with medical standards.
3. Regular internal quality audits and inspections are conducted
4. Opportunities for Automation and Digitization
 - Many repetitive tasks could be automated to improve consistency and speed.
 - Adoption of digital tools for SOP access, performance monitoring, and communication was limited but highly needed.
5. There are many free spaces present within the company premises. These could be used to expand production.
6. The company maintains updated documentation related to medical device regulations.
7. Lack of updated machinery.

7.2 CONCLUSION

The business process mapping study conducted at BPL Medical Technologies has provided a comprehensive understanding of the organization's core processes and operational workflows. By mapping processes, the study successfully highlighted key inefficiencies, redundancies, and performance gaps that may be affecting overall productivity.

Through the use of various tools such as flow analysis, fishbone diagrams, and a combination of qualitative and quantitative process analysis techniques, the study was able to identify root causes behind recurring issues such as high defect rates, delays in raw material supply from vendors, and communication breakdowns. It also revealed opportunities for automation, improved training, and more effective use of machinery and resources.

The proposed improvements, once implemented, are expected to bring measurable benefits such as enhanced workflow efficiency, reduction in error rates, improved compliance with medical device regulations, and better coordination across departments. These outcomes are especially crucial for a company like BPL Medical Technologies, which operates in a highly regulated and quality-sensitive industry.

In conclusion, this process mapping initiative not only serves as a foundation for immediate improvements but also sets the stage for a culture of continuous process optimization, innovation, and long-term operational excellence. Ongoing monitoring and regular reviews of these processes will be essential to ensure that the improvements remain effective and relevant in a dynamic business environment.

7.3 BIBLIOGRAPHY

- User manuals of products
- SOPs of departments
- <https://www.bplmedicaltechnologies.com/>

7.4 ABBREVIATIONS

- BPL – British Physical Laboratories
- SOP – Standard Operating Procedures
- PDCA – Plan Do Check Act
- IBP – Invasive Blood Pressure
- ECG – Electrocardiogram
- DVI – Digital Visual Interface
- HIS – Hospital Information System
- PM-JAY – Pradhan Mantri Jan Arogya Yojana
- EU MDR – European Union Medical Device Regulation